

ANSWER KEY — Integration Lesson 12 Test: Motion II — Acceleration & Gravity

For the grader · full working shown · give credit for correct method with small arithmetic slips · $g = 10 \text{ m/s}^2$ throughout

1. A sled starts from rest and accelerates at a steady 5 m/s^2 . Write $v(t)$, and find the velocity after 6 seconds.

Work: velocity = starting velocity + integral of acceleration: $v(t) = 0 + \int_0^t 5 \, du = 5t$. Then $v(6) = 5 \times 6 = 30$. Units: $(\text{m/s}^2) \times \text{s} = \text{m/s}$. **Answer: $v(t) = 5t$; $v(6) = 30 \text{ m/s}$.**

2. A skateboarder is already moving at 4 m/s and accelerates at 3 m/s^2 . Write $v(t)$ and find the velocity after 2 seconds.

Work: change in velocity = $\int_0^t 3 \, du = 3t$; add the start: $v(t) = 4 + 3t$. Then $v(2) = 4 + 6 = 10$. (Check $t = 0$: $v = 4$ ✓.) **Answer: $v(t) = 4 + 3t$; $v(2) = 10 \text{ m/s}$.** (Losing the starting 4 is the common error.)

3. A ball is dropped from a hot-air balloon. After 4 seconds, how fast is it moving and how far has it fallen?

Work: dropped means $v_0 = 0$. Integrating $a = 10$ gives $v = 10t$; integrating again gives distance fallen = $5t^2$. $v(4) = 40$; $s(4) = 5 \times 16 = 80$. **Answer: 40 m/s , having fallen 80 meters .**

4. A coin is dropped from a 45-meter rooftop. How long to hit the ground, and how fast when it lands?

Work: distance fallen = $5t^2$, so set $5t^2 = 45 \rightarrow t^2 = 9 \rightarrow t = 3 \text{ s}$. Landing speed: $v = 10t = 10 \times 3 = 30$. (Check: $5 \times 3^2 = 45$ ✓.) **Answer: 3 seconds , landing at 30 m/s .**

5. An object's velocity is $v(t) = 2 + 6t \text{ m/s}$. Use an integral to find how far it travels in the first 3 seconds.

Work: distance = $\int_0^3 (2 + 6t) \, dt = [2t + 3t^2]_0^3 = (6 + 27) - 0 = 33$. Units: $(\text{m/s}) \times \text{s} = \text{m}$. **Answer: 33 meters .**

6. A ball is thrown straight up at 40 m/s . Find the peak time, peak height, and landing time.

Work: $v(t) = 40 - 10t$. Peak when $v = 0$: $t = 4 \text{ s}$. Peak height = $\int_0^4 (40 - 10t) \, dt = [40t - 5t^2]_0^4 = 160 - 80 = 80 \text{ m}$. Lands when $s(t) = 40t - 5t^2 = 5t(8 - t) = 0 \rightarrow t = 8 \text{ s}$ (double the peak time, as expected). **Answer: peak at $t = 4 \text{ s}$, height 80 m , lands at $t = 8 \text{ s}$.**

7. A car moving at 30 m/s brakes with acceleration -6 m/s^2 . How long to stop, and how far while stopping?

Work: $v(t) = 30 - 6t$; stops when $30 - 6t = 0 \rightarrow t = 5 \text{ s}$. Distance $= \int_0^5 (30 - 6t) dt = [30t - 3t^2]_0^5 = 150 - 75 = 75 \text{ m}$. (Triangle check: $\frac{1}{2} \times 5 \times 30 = 75 \checkmark$.) **Answer: stops in 5 seconds, after 75 meters.**

8. Acceleration staircase: $a = 2$ from $t = 0$ to 3, then $a = 4$ from $t = 3$ to 5; starting velocity 5 m/s. Find $v(5)$.

Work: area under the acceleration graph = change in velocity. Rectangle 1: $2 \times 3 = 6 \text{ m/s}$. Rectangle 2: width $5 - 3 = 2$ (NOT 5), so $4 \times 2 = 8 \text{ m/s}$. Total change = 14; $v(5) = 5 + 14 = 19$. **Answer: 19 m/s.** (Common error: width 5 for the second rectangle $\rightarrow 26$, wrong; or forgetting the starting 5 $\rightarrow 14$, wrong.)

9. Derive $v = v_0 + at$ by integrating a constant acceleration a .

Work: acceleration is the rate of change of velocity, so the CHANGE in velocity from 0 to t is $\int_0^t a du = [au]_0^t = at$ (a rectangle of height a , width t). An integral only gives the change, not the actual velocity — the speedometer already read v_0 at the start — so the true velocity is start + change: $v = v_0 + at$. Full credit needs: the integral, the value at , and a reason for adding v_0 . **Answer: $v = v_0 + at$, with v_0 added because the integral gives only the CHANGE in velocity.**

10. Rocket from rest: $a = 8 \text{ m/s}^2$ for 3 s, then coasts at constant velocity for 5 s. Velocity after phase 1 and total distance over all 8 seconds?

Work: Phase 1: $v(t) = 8t \rightarrow v(3) = 24 \text{ m/s}$. Phase 1 distance $= \int_0^3 8t dt = [4t^2]_0^3 = 36 \text{ m}$. Phase 2: coasting means $a = 0$, velocity stays 24 m/s $\rightarrow$ distance $= 24 \times 5 = 120 \text{ m}$. Total $= 36 + 120 = 156 \text{ m}$. **Answer: 24 m/s after phase 1; total distance 156 meters.**